

# proxy\_forward\_benchmark.sh — user guide

**Revision: 1 Last modified:** 2026-07-01T13:10:00Z

## Overview

tests/benchmark/proxy\_forward\_benchmark.sh is the §11.4.169 **benchmarking / performance** test-type for the live HTTP forward proxy (Squid on 127.0.0.1:34128). It measures forward-proxy request **latency** (nearest-rank p50 / p95 / p99 + min / max / mean) and sequential **throughput** (successful requests per wall-second) by driving N (default 200) plain-HTTP requests through the proxy to a 204-returning endpoint and capturing **every** request's `%{http_code} + %{time_total}` to an evidence file. The verdict rests on the captured per-request samples, never a summary number (§11.4.6 no fake numbers, §11.4.69 captured evidence).

It is the performance sibling of the §11.4.85 stress suite (tests/stress/proxy\_forward\_stress.sh) and reuses the same anti-bluff verdict contract from tests/lib/evidence.sh.

## Prerequisites

- The base proxy stack UP (./start), Squid listening on :34128.
- bash, curl, awk, sort, grep on PATH.
- Network reachability to BENCH\_TARGET (default www.gstatic.com) — an external outage is classified as an honest SKIP, never a false FAIL.

## Usage examples

```
# default: 200 sequential requests, host-safety caps applied
GOMAXPROCS=2 nice -n 19 ionice -c 3 bash tests/benchmark/
    proxy_forward_benchmark.sh
```

```
# smaller/larger sample, custom proxy/target
BENCH_N=50 PROXY_ADDR=127.0.0.1:34128 \
```

```
BENCH_TARGET=http://www.gstatic.com/generate_204 \  
bash tests/benchmark/proxy_forward_benchmark.sh
```

Env overrides: PROXY\_ADDR (default 127.0.0.1:34128), BENCH\_TARGET, BENCH\_EXPECT (default 204), BENCH\_N (default 200), CURL\_MAX\_TIME (default 20), BENCH\_EVIDENCE\_DIR.

## Edge cases

- **Proxy dropped requests while target reachable directly** → OVERALL=FAIL (exit 1) — a real proxy defect (§11.4.68 no fail-open, never masked as SKIP).
- **Target unreachable via proxy AND directly, port listening** → SKIP:network\_unreachable\_external (exit 3) — third-party outage, not a proxy defect (§11.4.1 no false-FAIL of a healthy proxy).
- **Proxy port not listening, no direct signal** → SKIP:topology\_unsupported (exit 3) — the base stack is down (§11.4.3).
- A hung request cannot exceed CURL\_MAX\_TIME (bounds the whole run) and is recorded as 000.

## Internal behaviour

1. Sources tests/lib/evidence.sh (walks up to find the repo root).
2. Fires N sequential curl -x http://\$PROXY\_ADDR \$BENCH\_TARGET, appending %{time\_total} to a sample set and counting EXPECT responses.
3. Sorts the samples, computes nearest-rank percentiles + throughput, writes latency.txt (raw samples + distribution) and benchmark.evidence under qa-results/benchmark/proxy\_forward\_<ts>/.
4. Cross-checks a direct (non-proxied) probe to classify a shortfall as a proxy defect vs an external outage.
5. Emits one structured PASS ( $\geq$  N successful 204s) / FAIL / SKIP verdict via ab\_pass\_with\_evidence / ab\_skip\_with\_reason, citing latency.txt.
6. trap cleanup removes only the scratch dir (§11.4.14); evidence preserved.

## Last verified

2026-07-01 — live run against the base proxy: **200/200** proxied 204s, p50=0.086s p95=0.088s p99=0.091s (min 0.085 / max 0.102 / mean 0.087), throughput 10.841 req/s, OVERALL=PASS. Evidence: qa-results/benchmark/proxy\_forward\_20260701T130414Z/latency.txt.

## Related scripts

- `tests/stress/proxy_forward_stress.sh` — §11.4.85 stress sibling.
- `tests/lib/evidence.sh` — shared anti-bluff verdict helpers.
- `docs/design/hardening/Status.md` — §11.4.169 test-type coverage matrix.